Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Mathematical Sciences

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Daftar Isi

1 PENDAHULUAN **1**

2 TINJAUAN PUSTAKA **2**

1 Section 1 2

2 Acknowledgments 3

1 Section 1

The paper will look professional and is typed in LaTeX. Make sure the cover page includes the Project Title, Team Names, Date, and PIC Math Logo (1 point).

Remaining pages should include: (7 points)

1.Problem summary: state the industrial problem, its context, and its significance; include a list of most important questions to be addressed. State the industrial sponsor of the project.

2.Statement of your results, with assumptions and questions(one page).

3.Details about your (one of several) approaches to problem; elaborate which questions are important and why; your results and/or anticipations (this is where your mathematics details fit); relevant information from your sources (citing for example [2]) and what the team is incorporating from each source; details of your solution, if any; relevant tables and graphs (tables and graphs pages do not count towards the page limit). State your work and include references - you might never make use of reference [1] but it is a great paper.

4.Conclusions with questions, implications and ideas for future work

5.Extended Bibliography List (not included in page limit) in the following format:

[1] Author name(s). Paper title. Journal name, volume XY, pages ab-cd, year.

6.Acknowledgments page, stating:

"PIC Math is a program of the Mathematical Association of America (MAA) and the Society for Industrial and AppliedMathematics (SIAM). Support is provided by the National Science Foundation (NSF grant DMS-1345499)."

7.Advisor names, etc.

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2 Acknowledgments

PIC Math is a program of the Mathematical Association of America (MAA) and the Society for Industrial and Applied Mathematics (SIAM). Support is provided by the National Science Foundation (NSF grant DMS-1345499).

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[3] K. Groechenig, *Foundations of time-frequency analysis*, Birkhäuser Boston, 2001.